

Personal Project Related Work &
Technical Background
Minor AI for Society

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1. Document History

Table 1. Document History

Version	Date	Status	Author	Description
0.1	18-03-2025	Initial Version	Casey Stijlaart	Related Work
1.0	19-03	First Version	Casey Stijlaart	Completed First Version

2. Introduction

This document serves as a research report for the related work and technical background of the personal project for the Minor AI for Society course. The purpose of this report is to provide an overview of the related work in the field of AI and its applications in society, as well as the technical background necessary for the development of the project. The report is structured into two main sections: Related Work and Technical Background. The Related Work section will review existing research and applications of AI in society, while the Technical Background section will cover the fundamental concepts and technologies that underpin the project.

3. Related Work

This section reviews existing research and commercial solutions related to plant health monitoring systems, with a focus on IoT-based approaches, AI/ML techniques, and the integration of these technologies in practical applications. The review is structured into three main categories: IoT-Based Plant Monitoring Systems, AI/ML-Based Plant Health Monitoring Systems, and Commercial Solutions. Each category highlights key studies, methodologies, and findings that inform the design and development of the proposed system. The section concludes with a summary of the existing systems and their relevance to the project.

3.1. IoT-Based Plant Monitoring Systems

Currently, a variety of plant monitoring systems are available, most of which have been developed by students, hobbyists, or small-scale developers. These systems generally focus on environmental monitoring rather than advanced AI-based analysis.

Many of these solutions are open-source, providing access to hardware and software designs, while others are proprietary. The majority of existing systems do not integrate AI or Machine Learning; instead, they monitor environmental parameters—such as soil moisture, temperature, and humidity—and notify users when predefined thresholds are exceeded.

Some implementations are more advanced, incorporating automated irrigation systems that water plants when certain conditions are met, improving plant care efficiency. Other systems are simpler, only sending alerts or notifications to the user without any automation.

Examples of existing systems include:

- Plant Monitoring System by Shishir3D []
- IoT-based Plant Health Care System (MDPI) []
- IoT-Based Plant Health Care System (Academia.edu) []
- IoT-Based Smart Irrigation System (Academia.edu) []

3.2. AI/ML-Based Plant Health Monitoring Systems

In contrast to traditional IoT-based plant monitoring systems, recent developments have begun integrating Artificial Intelligence (AI) and Machine Learning (ML) techniques. These systems are capable of not only monitoring environmental parameters but also analyzing data to make intelligent predictions, detect anomalies, and optimize plant care.

AI/ML-based plant health monitoring systems can:

- Predict irrigation needs based on soil moisture trends and weather conditions.
- Detect early signs of plant stress or disease using sensor data or images.
- Provide automated recommendations or interventions to improve plant health.
- Learn and adapt to the specific requirements of different plant species over time.

These advancements enable more proactive and efficient plant management, surpassing the reactive threshold-based alerts of traditional systems.

Examples of AI/ML-based systems include:

- IoT-Based Smart Plant Monitoring System (Emergent Mind) []
- AI-Enabled IoT-Based Smart Indoor Plant Care System (Academia.edu) []
- Deep Learning for Plant Health Monitoring (Nature Scientific Reports) []
- AI-Driven Smart Plant Monitoring (IJSAT Journal) []
- Plant Health Monitoring System (GitHub) []

These systems illustrate the potential for combining IoT hardware with AI/ML algorithms to achieve intelligent, autonomous plant care and monitoring.

These existing approaches highlight the potential for enhancement by integrating AI and ML techniques, which could enable predictive care, anomaly detection, and intelligent decision-making for plant health and irrigation management.

3.3. Commercial Solutions

Several commercial solutions exist for crop and plant monitoring; however, most of these systems are designed for large-scale agricultural applications rather than small-scale or individual plant care. These solutions often integrate advanced sensors and IoT platforms to monitor soil conditions, plant health, and environmental factors at scale.

While they provide reliable data and automated management for farms and commercial crops, their focus and cost make them less suitable for typical household or small-scale plant monitoring projects. Nonetheless, these commercial systems demonstrate the potential of IoT and sensor integration for predictive and automated plant care.

Examples of commercial systems include:

- Connected Green – Soil Moisture Sensor []: A commercial sensor solution for monitoring soil moisture in large crop environments. Available at: <https://connectedgreen.nl/bodemvochtsensor>
- Cropwise Interra Scan []: A precision agriculture platform that uses IoT devices and analytics for crop monitoring and management. Available at: <https://www.cropwise.com/netherlands/interra-scan>
- Vivent Biosignals – Food and Agriculture Inputs []: Provides sensor-based monitoring and analytics for plant and crop health on a commercial scale. Available at: <https://vivent-biosignals.com/solutions/food-and-ag-inputs>

These systems illustrate how IoT and sensor technologies are applied in commercial agriculture, but also highlight the gap for small-scale, AI/ML-based plant care systems suitable for personal or hobbyist use.

3.4. Comparison of Existing Systems

Table 2. Comparison Table of Plant Monitoring Systems

Feature / System Type	IoT-Based (Non-AI)	AI/ML-Based	Commercial Solutions
Typical Cost	Low (€5–50)	Low–Medium (Research prototypes €20–100)	High (€100+ per sensor / system)
Target Use	Hobbyist / student projects	Research / experimental	Commercial crops / large-scale agriculture
Automation	Some (watering, notifications)	Advanced (predictive watering, alerts, recommendations)	Advanced (field irrigation, crop management)
AI/ML Integration	No	Yes (predictive and adaptive)	Limited (mostly analytics for large-scale optimization)
Sensors Used	Basic environmental (soil moisture, temperature, humidity)	Environmental + image/spectral data	Advanced multi-sensor arrays, remote sensing
Open Source Availability	Often open source	Mostly research / prototype	Rarely open source
Scalability	Easy for small-scale deployment	Moderate, experimental	High for large fields, not small-scale
Gaps Identified	Lacks AI/ML, predictive capabilities	Limited accessibility and cost-effective deployment	Not suitable for indoor/home plants, high cost

3.5. Summary of Existing Plant Monitoring Systems

Plant monitoring systems can be broadly categorized into three main groups:

- **IoT-Based Systems (Non-AI/ML):** These systems monitor environmental parameters such as soil moisture, temperature, and humidity, often sending alerts when thresholds are exceeded. Some include basic automation, like watering when soil moisture is low. They are widely implemented in open-source hobbyist projects and student prototypes. However, they lack AI/ML capabilities for predictive or adaptive care.
- **AI/ML-Based Systems:** Recent research and prototype systems integrate AI/ML to predict irrigation needs, detect plant stress or disease, and provide intelligent recommendations. These systems are still largely experimental or research-based, with limited adoption in consumer-grade products. They demonstrate the potential for proactive, adaptive plant care that goes beyond threshold-based monitoring.
- **Commercial Solutions:** Commercial products primarily target large-scale agriculture, monitoring crop health across fields using sophisticated sensors and IoT platforms. These solutions are highly capable but are not designed for regular indoor or household plants, and they are often cost-prohibitive for small-scale applications.

Despite these developments, there is a clear gap for affordable, small-scale AI/ML-based plant health monitoring systems suitable for hobbyists or home users. Most current systems either:

1. Focus on simple threshold-based alerts (IoT-only).
2. Are experimental research prototypes (AI/ML).
3. Are expensive and targeted at commercial agriculture.

There is no widely-available, low-cost solution that combines AI/ML intelligence with IoT for personal plant care.

4. Technical Background

This section provides an in-depth technical background relevant to the design and implementation of the plant health monitoring system. It covers various AI/ML approaches, datasets, model research, embedded system constraints, validation strategies, and design implications that inform the development of the project.

4.1. Embedded Systems for Edge AI

4.1.1. Low-Power Microcontrollers

Low-power microcontrollers enable embedded artificial intelligence systems by combining compact size with energy-efficient computation. These devices can interface directly with environmental sensors while performing real-time inference locally, which is critical for applications such as autonomous plant monitoring. Research and community projects have demonstrated that even constrained architectures like the ESP32 family can execute lightweight machine learning workloads when models are optimized for on-device inference [plantmonitoring_github][plantmonitoring_mdpi][planthealth_academia].

4.1.2. TinyML Concept

Tiny Machine Learning (TinyML) refers to deploying optimized machine learning models on microcontrollers and other resource-constrained embedded devices. TinyML workflows focus on converting trained models (e.g., small neural networks or classical classifiers) into lightweight, quantized representations that can run efficiently with limited RAM and processing power [ai_plant_academia][ai_plant_emergentmind].

Key aspects of TinyML include: * **Quantization:** Reducing model weights and activations to 8-bit or lower precision. * **Memory efficiency:** Ensuring models fit within small flash and SRAM capacities. * **Optimization:** Tailoring model architecture for real-time inference on low-power hardware.

In plant monitoring systems, TinyML enables: * Real-time event detection such as soil moisture thresholds or plant stress. * Minimal dependency on cloud infrastructure, reducing latency and preserving privacy. * Energy-efficient operation, allowing prolonged deployment in remote or battery-powered environments.

4.1.3. Edge AI vs Hybrid AI

Edge AI and hybrid AI represent different approaches for deploying machine learning in embedded systems:

- **Edge AI:** All inference occurs locally on the device, with data captured, processed, and classified in real-time. Advantages include low latency, independence from network connectivity, and improved privacy. This is particularly suitable for plant monitoring systems operating in greenhouses or outdoor areas where connectivity is limited [ai_plant_nature][ai_plant_ijsat].
- **Hybrid AI:** Combines local edge inference with cloud-based processing. Lightweight models handle immediate tasks on-device, while complex analytics or model retraining occurs in the

cloud. This approach allows more sophisticated analysis but relies on reliable connectivity and may introduce latency.

For autonomous plant monitoring, **Edge AI** is generally preferred for resilience, energy efficiency, and responsiveness, while hybrid AI may support periodic model updates or long-term analytics [ai_plant_github][commercial_connectedgreen].

4.1.4. Edge AI on ESP32

Although documentation specific to the ESP32 CH2102 is limited, several resources demonstrate the feasibility of Edge AI on ESP32-class microcontrollers:

- **Edge Impulse on ESP32:** Provides end-to-end TinyML workflows including data acquisition, training, and deployment to ESP32 boards for tasks such as vision and sensor-based classification [ai_plant_emergentmind][ai_plant_academia].
- **Community projects:** Open-source projects show TensorFlow Lite Micro and Edge Impulse models running on ESP32 devices for plant health monitoring, demonstrating real-time on-device inference and sensor integration [plantmonitoring_github][ai_plant_github].
- **Research applications:** ESP32 microcontrollers have been used in IoT-based smart irrigation and plant monitoring systems where lightweight ML models predict watering needs or detect environmental stress [planthealth_academia][smartirrigation_academia].

These examples collectively indicate that Edge AI is practical and effective on ESP32-class MCUs, including variants like the CH2102.

4.1.5. Edge AI Workflow

The workflow of how Edge AI operates in practice is shown in the diagram below. It illustrates the process from sensor data acquisition to on-device inference and actuator/alert response.

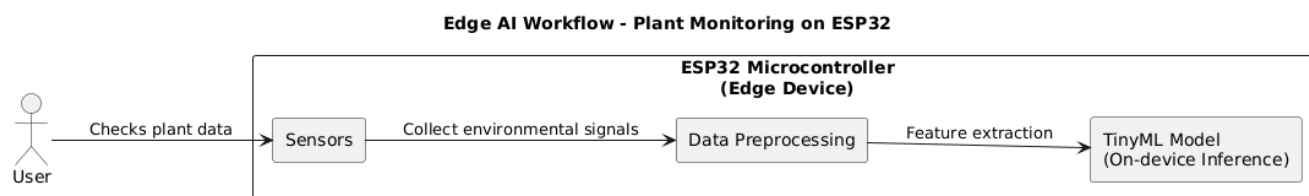


Figure 1. Edge AI Workflow

4.2. Sensor Data Analysis

In a plant monitoring system, sensor data is the foundation of all decision-making. Efficient analysis of this data on low-power microcontrollers requires careful consideration of sensor types, preprocessing techniques, and anomaly detection strategies.

4.2.1. Types of Sensor Data

Plant monitoring systems typically rely on multiple environmental and plant-specific sensors, such as:

- **Soil Moisture Sensors:** Measure volumetric water content to determine irrigation needs.
- **Temperature and Humidity Sensors:** Track ambient conditions that affect plant growth.
- **Light Sensors (e.g., PAR or LDR):** Assess sunlight exposure and photosynthetic potential.
- **Additional sensors:** CO₂ concentration, pH, nutrient levels, or plant stress indicators can be added depending on system requirements.

Each sensor produces data streams that are generally time-series in nature, making temporal patterns important for detecting anomalies or triggering automated responses.

4.2.2. Time-Series Anomaly Detection

Time-series anomaly detection is critical in plant monitoring for identifying abnormal conditions such as drought stress, overwatering, or environmental irregularities. On low-power devices like the ESP32, lightweight algorithms are preferred:

- **Statistical Methods:** Z-score, moving average deviations, or seasonal trend decomposition.
- **Machine Learning Methods:** Small neural networks or autoencoders trained to detect deviations from normal patterns.
- **Threshold-Based Alerts:** Simple, rule-based triggers suitable for on-device real-time monitoring.

By detecting anomalies locally, the system can act accordingly without relying on cloud connectivity.

4.2.3. Lightweight Preprocessing

Preprocessing sensor data before model inference is essential on constrained devices to reduce memory and computation requirements. Techniques include:

- **Normalization and Scaling:** Adjusting sensor values to a standard range suitable for ML models.
- **Noise Filtering:** Using low-pass filters or moving averages to remove transient spikes.
- **Feature Extraction:** Calculating simple features such as rolling means, variance, or delta changes to summarize sensor signals.
- **Dimensionality Reduction:** Combining correlated sensor readings or reducing sampling frequency to decrease data size.

Efficient preprocessing ensures that the TinyML model receives clean, compact, and informative input, enabling accurate predictions while conserving ESP32 resources.

4.3. Datasets

4.3.1. Public Sensor Datasets

4.3.2. Data Collection Considerations

4.3.3. Limitations

4.4. Model Overview

4.4.1. Lightweight ML Models

4.4.2. Time-series Prediction Basics

4.4.3. Suitability For ESP32/TinyML

4.5. Validation Overview

4.5.1. Basic Dataset Splitting

4.5.2. Evaluation Metrics

4.5.3. Performance Considerations

4.6. Summary of Technical Background

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